

Benjamin Kim

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Education

University of Southern California,
Viterbi School of Engineering
Bachelor of Science, Computer Science

Los Angeles, CA
Expected May 2029
GPA 3.92

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript, Swift, C#, HTML/CSS

Technologies: MySQL, PROC SQL, SQLite, Jupyter, Data Analysis, Google Cloud Platform, AWS, Git, DynamoDB, Docker, Django

Experience

Figwork - Software Engineer Intern

July 2026 - Present

Recruitment Matching and Onboarding Facilitator app

- Engineered a large-scale job discovery and scraping pipeline spanning 60+ ATS platforms, combining localized search, CT/DNS discovery, ATS fingerprinting, and optimized API orchestration to collect 600,000+ job postings.
- Evaluated performances of OCR models by automating video-based testing pipelines, then processing and analyzing the results end-to-end. Provided failure cases, as well as efficiency and accuracy of each model combined with their pricing.
- Configured official assets in the company website to be processed separately depending on the browser and the operating system. Audited the full stack structures for reliability and routed residential proxy, JS renderer, and API/vendors.

Gray - Full Stack Developer

May 2026 - Present

- Built a privacy-preserving macOS workflow agent that records user interactions across applications, which learns reusable workflows and converts them into replayable automations using TypeScript, Electron, JXA, and macOS Accessibility APIs.
- Engineered a real-time capture and sanitization pipeline for user input, application, clipboard, and UI elements, with secure-field detection. Included multi stage secret redaction, clipboard hashing, and sparse local only visual context.
- Developed an LLM pipeline using OpenAI Structured Outputs and Zod to transform recorded activity into validated automation scripts. Established workflow to convert hallucinated or unsupported actions into safe manual steps for the user.
- Built a fault-tolerant JXA automation runtime that verifies each action against live UI, re-anchors clicks to the current window, and automates retries for failed steps before holding for review. Added supervised pause, skip, and takeover controls with destination scope and commit gates so that the users can approve uncertain or critical actions.
- Architected the Electron client around isolated main, preload, and renderer processes with typed IPC. Built the React interface around a state machine that transitions between each stage of recording and evaluation. Implemented permission onboarding and local-only library to optimize data storage and utilization.

USC Rocket Propulsion Lab - Full Stack Developer

January 2026 - Present

- Built automatic inventory management system that enables tracking all items required by the lab by implementing sorting mechanisms and parsing csv files for immediate lookups. Included authentication, architecture, and security development.
- Established locating service with quick database lookup to automatically guide lab members to the position of each material.

Smart Radar System (Radar technologies company) - Data Science Intern

June 2024 - August 2024

- Utilized neural networks, deep learning, transformers, radar systems, and sensor fusing and studied about the development of radars and software. Learned to analyze data and differentiate radar readings for bicycles, motorcycles, and cars.

Selected Projects

Rewind Rescue XR

April 2026

- Immersive mixed-reality emergency response trainer built for real-world training impact. Built using XR stack, Unity, and C# to simulate a 360° environment with user tracking to pressurize them while they are performing first aid.

StepDue, Task Focus Extension

February 2026

- Designed StepDue, a browser-native accountability tool for workers that provides versatile focus metrics via session tracking, progress visualization, and optional distraction controls.

NewUrl, AI-Assisted URL Shortener

December 2025

- Developed a scalable full-stack URL shortening service with AI-generated aliases and QR code creation using Node.js and Redis-based key-value storage.

Markdown Notetaker, Full-Stack Application

November 2025

- Built a full-stack real-time Markdown note-taking application using React and Spring Boot with live preview, REST APIs, and file export functionality.